

Geometric monotones of violations of quantum realism

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Quantum realism, as introduced by Bilobran and Angelo [*Europhys. Lett.* **112**, 40005 (2015)], states that projective measurements in quantum systems establish the reality of physical properties, even in the absence of a revealed outcome. This framework provides a nuanced perspective on the distinction between classical and quantum notions of realism, emphasizing the contextuality and complementarity inherent in quantum systems. While prior works quantified violations of quantum realism (VQRs) using measures based on entropic distances, here we extend the theoretical framework to geometric distances. Building on an informational approach, we derive geometric monotones of VQRs using trace distance, Hilbert-Schmidt distance, Schatten p distances, and Bures and Hellinger distances. We identify Bures and Hellinger distances as uniquely satisfying all minimal criteria for a bona fide VQR monotone. Remarkably, these distances can be expressed in terms of symmetric Rényi and sandwiched Rényi divergences, aligning geometric and entropic approaches. Our findings suggest that the realism-information relation implies a deep connection between geometric and entropic frameworks, with only those geometric distances expressible as entropic quantities qualifying as valid monotones of VQRs. This work highlights the theoretical and practical advantages of geometric distances, particularly in contexts where computational simplicity or symmetry is important.

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I. INTRODUCTION

The concept of quantum realism of observables, as proposed by Bilobran and Angelo [1], can be summarized as follows. After the projective measurement of a physical property in a quantum system, we can assert that the property is now well-defined, i.e., it is real for the state of the system. Any subsequent projective measurement of the same property will yield the same outcome because the new measurement merely reveals a preexisting *state of affairs*—or reality. This reasoning also holds for nonselective (or nonrevealed) projective measurements. Even if the experimentalist is not present in the laboratory when the measurement occurs or forgets the result, the measurement still takes place, establishing the property as real. Thus, a projective measurement, even without a revealed outcome, serves as the mechanism by which realism emerges in a quantum system.

Consider, for instance, the measurement of the observable $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$ on the state $|\psi\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. After the measurement, the system's state will collapse to either $|0\rangle$ or $|1\rangle$. At this point, the realism (or reality) of σ_z is established, meaning that any subsequent measurement of σ_z will yield the same outcome. In the case of a nonselective measurement, however, the postmeasurement state of the system would be the mixed state $|0\rangle\langle 0| + |1\rangle\langle 1|$. This mixed state indicates that the system is definitely in either $|0\rangle$ or $|1\rangle$, but the observer lacks knowledge of which one. This lack of knowledge is subjective and classical—it reflects the observer's ignorance rather than an intrinsic uncertainty in the quantum state, as is

the case with superpositions. By contrast, the observable σ_x is real for the initial state $|\psi\rangle$ because the state is an eigenstate of σ_x , with a well-defined eigenvalue. This highlights the fundamental difference between realism in quantum mechanics and classical realism. Classical realism assumes that every physical property of a system is well-defined at all times, independent of observation or the presence of an observer. In contrast, quantum realism recognizes that certain pairs of incompatible observables, such as σ_x and σ_z , cannot always be simultaneously well-defined. This limitation is intrinsic to the nature of quantum systems and reflects the contextuality and complementarity that are hallmarks of quantum theory.

The situation changes dramatically when we consider bipartite systems. If Alice and Bob share a maximally entangled pair of spin-1/2 particles, their local states are maximally mixed. Consequently, the spin observable for, say, Alice's particle is *locally* an element of physical reality, as a nonselective measurement of the identity operator does not alter the state. However, when considering the *global* description of the pair, any local nonselective measurement perturbs the global state. This demonstrates that the state of the pair violates quantum realism for the observable associated with either particle. This phenomenon diverges sharply from the Einstein-Podolsky-Rosen (EPR) notion of elements of physical reality [2], as applied to bipartite entangled states. According to EPR, local measurements should not influence the state of affairs at remote locations.

A method to quantify violations of quantum realism (VQRs) was proposed in [1], where the authors referred to VQRs as *irrealism*. They introduced an entropic distance between the measured state and the original state as a means of quantification. These measures of VQRs were largely

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explored in theoretical [3–15] and experimental [16–20] approaches. Building on the informational framework of [3] and the experimental findings of [16,17], the authors of [21] presented a set of physically motivated axioms for monotones and measures of VQRs, where the latter requires more conditions than the former. In their work, they defined a notion of conditional quantum information—the complement of quantum conditional entropy—using general relative entropies, also called divergences. They further showed how variations in conditional information can be used to derive expressions for VQR measures. Their argument relies on a *realism-information relation*: Since nonselective projective measurements are responsible for the emergence of realism, the increase in conditional information about the system, as encoded in the environment, could serve as a monotone of VQRs. This connection is formalized using the Stinespring dilation theorem [22].

Contributions

In this work, we extend the concept of conditional information based on relative entropies from [21] to geometric distances and successfully derive geometric monotones of VQRs. Geometric distances, or simply *distances*, share properties with relative entropies, such as contraction under completely positive trace-preserving (CPTP) maps, invariance under unitary transformations, and joint convexity. Additionally, distances are inherently symmetric and satisfy the triangle inequality, making them advantageous in certain contexts. Moreover, some distances are computationally simpler to evaluate, offering a practical benefit over entropic measures. Although none of the distances we studied are additive, making them not good candidates for measures of VQRs, some of them can still be classified as monotones of VQRs. Among the distances we explore, we have trace distance, the Hilbert-Schmidt distance [22], Schatten p distances [23,24], and Bures [24,25] and Hellinger [24,26] distances. We found out that, among these distances, Bures and Hellinger distances are the only candidates that satisfy all the minimal requirements for a bona fide monotone of VQRs. Trace distance, for instance, provides a quantifier that fails to identify VQRs for some states. The Hilbert-Schmidt distance, on the other hand, provides a quantifier of VQRs that decreases with the addition of completely uncorrelated states. Interestingly, Bures and Hellinger distances can be written as functions of Rényi and sandwiched Rényi divergences in the particular cases where these divergences are symmetric. Our results suggest that, if one assumes the realism-information relation, then only the geometric distances that can be framed as entropic quantities are suitable for deriving monotones of VQR.

II. FUNDAMENTAL CONCEPTS

A. Violations of quantum realism

Let us mathematically formalize the concept of quantum realism presented in the Introduction by following Refs. [1,3,21]. Let $A = \sum_a a A_a$ be a d -output observable acting on $\mathcal{H}_A$, where $A_a = |a\rangle\langle a|$ are projectors satisfying $\sum_a A_a = \mathbb{1}_A$. Additionally, define the nonselective projective measurement of the observable A as $\Phi_A(\rho_{AB}) = \sum_a (A_a \otimes \mathbb{1}_B) \rho_{AB} (A_a \otimes \mathbb{1}_B)$, where ρ_{AB} is the state of a bipartite system in $\mathcal{H}_A \otimes \mathcal{H}_B$. We say that ρ_{AB} has realism defined for A iff $\Phi_A(\rho_{AB}) = \rho_{AB}$. States that already possess realism for a particular observable are invariant under a nonselective (or nonrevealed) projective measurement of that observable, i.e., $\Phi_A(\Phi_A(\rho)) = \Phi_A(\rho)$. Henceforth, the *violation of the quantum realism of A given ρ* is defined as

$$\mathfrak{I}_A(\rho) := S(\Phi_A(\rho)) - S(\rho), \quad (1)$$

which can also be called the *irrealism* of A given ρ , and $S(\rho) := -\text{Tr}(\rho \log_2 \rho)$ is the von Neumann entropy. Moreover, we can decompose irrealism as the sum of local coherence and nonoptimized quantum discord, that is, $\mathfrak{I}_A(\rho) = C(\rho_A) + D_A(\rho)$, where $C(\rho_A) = \mathfrak{I}_A(\text{Tr}_B \rho)$, $D_A(\rho) = I(\mathcal{A} : \mathcal{B})_\rho - I(\mathcal{A} : \mathcal{B})_{\Phi_A(\rho)}$, and $I(\mathcal{A} : \mathcal{B})_\rho = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$ is the quantum mutual information. Consequently, correlations between parties $\mathcal{A}$ and $\mathcal{B}$ prevent the existence of elements of physical reality for observables on both parties. That is, any positive value of $D_{A(B)}(\rho)$ implies nonzero irrealism $\mathfrak{I}_{A(B)}(\rho)$.

Functionals such as (1) were also introduced in other works but with different names and different meanings, such as measures of wavelike information [27] and quantum incoherent relative entropy [28]. Also, if observable A is given by the spectral decomposition of ρ_A , then the realism condition $\Phi_A(\rho_{AB}) = \rho_{AB}$ is equivalent to definition of semiquantum states [29].

B. Quantifying violations of realism using quantum conditional information

Because the shape of (1) was chosen *ad hoc*, one could wonder if other entropic distances would be a better fit to quantify VQR. To answer that question, an axiomatic and physically motivated approach was taken in Ref. [21] to propose that irrealism should decrease in the presence of a flux of information from the system to the environment. Since measurements are the key elements by which a physical property gets defined, one could argue that quantifying how much information the environment $\mathcal{E}$ gets about the system $\mathcal{S}$ makes it possible to quantify how irrealism decreases. The figure of merit in this case is the complement of quantum conditional entropy

$$H(\mathcal{E}|\mathcal{S})_\Omega := S(\Omega) - S(\Omega_{\mathcal{E}}), \quad (2)$$

that is, the quantum conditional information [21]

$$I(\mathcal{E}|\mathcal{S})_\Omega := \ln d_{\mathcal{E}} - H(\mathcal{E}|\mathcal{S})_\Omega, \quad (3)$$

where Ω acts over $\mathcal{H}_S \otimes \mathcal{H}_E$, $\Omega_E = \text{Tr}_S \Omega$, and $d_E = \dim \mathcal{H}_E$. While the former measures the ignorance that we still have about $\mathcal{E}$ given that we know $\mathcal{S}$, the latter measures the knowledge (or information) we can obtain by accessing $\mathcal{E}$ conditioned on the information codified in $\mathcal{S}$. Note that $0 \leq I(\mathcal{E}|\mathcal{S})_\rho \leq \ln d_{\mathcal{E}|\mathcal{S}}$.

To connect conditional information and quantum realism, the authors in [21] considered a system in the state ρ_S that interacts with an environment in the state $|0\rangle\langle 0|$ such that $\Omega_0 := \rho_S \otimes |0\rangle\langle 0|$ and $\Omega_t := U_{SE} \Omega_0 U_{SE}^\dagger$ are the global states before and after the interaction U_{SE} . This interaction must result in a nonselective projective measurement over the system

after we trace out the environment, that is,

$$\text{Tr}_{\mathcal{E}}[U_{\mathcal{SE}}(\rho_S \otimes |0\rangle\langle 0|)U_{\mathcal{SE}}^\dagger] = \Phi_A(\rho_S), \quad (4)$$

where A is some observable of interest (see the previous section). Equation (4) guarantees that the information about the observable A in $\mathcal{S}$ that is going to be codified in $\mathcal{E}$ —through the unitary interaction $U_{\mathcal{SE}}$ —is associated with the emergence of realism about A in the system through the map Φ_A . The authors defined that this increment of realism about A given ρ , denoted by $\mathfrak{R}_A(\rho)$, should vary such that

$$\Delta \mathfrak{R}_A(t) \equiv \Delta I(\mathcal{E}|\mathcal{S})_{\Omega_0 \rightarrow \Omega_t}, \quad (5)$$

which can be thought of as a *realism-information relation*, where $\Delta \mathfrak{R}_A(t) := \mathfrak{R}_A(\rho_S(t)) - \mathfrak{R}_A(\rho_S(0))$, with $\rho_S(0) := \rho_S$ and $\rho_S(t) := \text{Tr}_{\mathcal{E}}(\Omega_t) = \Phi_A(\rho_S)$. Additionally, the corresponding change in the environmental conditional information is $\Delta I(\mathcal{E}|\mathcal{S})_{\Omega_0 \rightarrow \Omega_t} := I(\mathcal{E}|\mathcal{S})_{\Omega_t} - I(\mathcal{E}|\mathcal{S})_{\Omega_0}$. In [21], the authors proved that relation (5) results in

$$\mathfrak{R}_A(\rho_S) = \ln d_A - S(\rho_S || \Phi_A(\rho_S)) = \ln d_A - \mathfrak{I}_A(\rho_S) \quad (6)$$

since the von Neumann relative entropy [22]

$$S(\rho || \sigma) := \text{Tr}(\rho \ln \rho - \rho \ln \sigma) \quad (7)$$

satisfies $S(\rho_S || \Phi_A(\rho_S)) = S(\Phi_A(\rho_S)) - S(\rho_S)$ [7]. Equation (6) shows that, by taking an informational approach to the problem of the emergence of realism through nonselective projective measurements, we can regain the expression for irreality of observables (1).

In addition to the realism-information relation (5), the authors in [21] also introduced a list of axioms that should be satisfied by any *bona fide monotone* of VQR.

Axiom 1. Realism and measurements. The realism of A acting on $\mathcal{H}_{\mathcal{A}}$ given ρ acting on $\mathcal{H}_{\mathcal{A}} \otimes \mathcal{H}_{\mathcal{B}}$, denoted by $\mathfrak{R}_A(\rho)$, is a non-negative real number and increases upon nonselective projective measurements of A , reaching a maximum when $\rho = \Phi_A(\rho)$, that is, $0 \leq \mathfrak{R}_A(\rho) \leq \mathfrak{R}_A(M_A^\epsilon(\rho)) \leq \mathfrak{R}_A(\Phi_A(\rho)) = \mathfrak{R}_A^{\max}$, where $M_A^\epsilon(\rho) := (1 - \epsilon)\rho + \epsilon\Phi_A(\rho)$ is the monitoring map.

Axiom 2. Role of other parts. (a) Discarding subsystems does not reduce realism, i.e., $\mathfrak{R}_A(\text{Tr}_{\mathcal{X}}(\rho)) \geq \mathfrak{R}_A(\rho)$ for $\mathcal{H}_{\mathcal{X}} \subseteq \mathcal{H}_{\mathcal{B}}$, and (b) adding an uncorrelated system does not change realism, i.e., $\mathfrak{R}_A(\rho \otimes \sigma) = \mathfrak{R}_A(\rho)$.

Axiom 3. Uncertainty relation. Two observables X and Y acting on $\mathcal{H}_{\mathcal{A}}$ cannot be simultaneous elements of reality in general, that is, $\mathfrak{R}_X(\rho) + \mathfrak{R}_Y(\rho) \leq 2\mathfrak{R}_A^{\max}$. Equality must hold only when $[X, Y] = 0$ or $\rho = \mathbb{1}_{\mathcal{A}}/d_{\mathcal{A}} \otimes \rho_{\mathcal{B}}$.

Axiom 4. Mixing. For a probabilistic ensemble $\{p_i, \rho_i\}$, realism is concave, i.e., $\mathfrak{R}_A(\sum_i p_i \rho_i) \geq \sum_i p_i \mathfrak{R}_A(\rho_i)$.

Moreover, if a quantifier of the VQR additionally satisfies additivity and flag additivity, it is called a *measure* of the VQR. Since none of the distances we studied are additive, only monotones of VQRs were found.

C. Alternative ways of quantifying quantum conditional information

An alternative way of expressing quantum conditional entropy is by means of divergences [30], with von Neumann relative entropy being the most prominent example. From (2)

and (7) it can be shown that

$$H(\mathcal{E}|\mathcal{S})_{\Omega} = -S(\Omega || \Omega_{\mathcal{S}} \otimes \mathbb{1}_{\mathcal{E}}). \quad (8)$$

On the other hand, the quantum conditional information (3) can be conveniently rewritten in terms of (7) as

$$I(\mathcal{E}|\mathcal{S})_{\Omega} = S\left(\Omega \left\| \Omega_{\mathcal{S}} \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}}\right.\right). \quad (9)$$

To better understand the meaning of conditional information, one can check that it can be decomposed as

$$I(\mathcal{E}|\mathcal{S})_{\Omega} = I(\Omega_{\mathcal{E}}) + I(\mathcal{E} : \mathcal{S})_{\Omega}, \quad (10)$$

where $I(\rho_{\mathcal{E}}) := S(\Omega_{\mathcal{E}} || \mathbb{1}_{\mathcal{E}}/d_{\mathcal{E}}) = \ln d_{\mathcal{E}} - S(\Omega_{\mathcal{E}})$ is the informational content of part $\mathcal{E}$ (because it is the complement of the entropy of $\mathcal{E}$) and $I(\mathcal{E} : \mathcal{S})_{\Omega} = S(\rho || \rho_{\mathcal{S}} \otimes \rho_{\mathcal{E}})$ is the quantum mutual information now written as a divergence. Since the mutual information is a measure of total correlations between the parts, the conditional information can be said to be composed of “local” and “global” information [31]. In the case of a product state such as $\rho_{\mathcal{S}} \otimes \sigma_{\mathcal{E}}$, the conditional information will be codified only in part $\mathcal{E}$. However, if the global state is maximally entangled, all the conditional information will be due to the correlations since the local state assigned to part $\mathcal{E}$ is the maximally mixed one.

It was observed in [30] and later expanded upon in [21] within the context of irreality measures that if one intends to explore alternative formulations of conditional entropy using other entropic measures, such as Rényi and Tsallis entropies, the approach outlined in (2) proves insufficient. This insufficiency arises due to the failure of data-processing inequalities. The same does not apply to approaches using divergences such as the one employed in (8). By replacing the von Neumann relative entropy $S(\cdot || \cdot)$ with other divergence measures such as sandwiched Rényi divergence [30] and Tsallis [32] relative entropy, we can construct alternative entropic measures for conditional information that still reach (10). Incidentally, these other measures for conditional information result in alternative ways to measure the quantum realism of observables through the realism-information relation (5) as extensively explored in [21].

D. Geometric distances and norms

In this paper, we explore the possibility of using geometrical distances to quantify conditional information—and consequently quantum realism—as an alternative to entropic measures. In what follows, we revisit the definition of *distance* and list the examples that we are interested in exploring, as well as their fundamental properties [33].

A *distance* is a function $d : M \times M \mapsto \mathbb{R}$ that maps each pair $x, y \in M$ onto a real number $d(x, y)$ satisfying *positive definiteness* $d(x, y) \geq 0$, where the equality holds iff $x = y$; *symmetry* $d(x, y) = d(y, x)$; and the *triangle inequality* $d(x, z) \leq d(x, y) + d(y, z)$. Positive definiteness is essential to obtain proper distinguishing measures. In addition, other properties are necessary for such measures to have a physical meaning. In the context of density states ρ and σ , a distance is said to be *contractive* under a CPTP map Λ iff

$$d(\Lambda(\rho), \Lambda(\sigma)) \leq d(\rho, \sigma). \quad (11)$$

In particular, contractive distances are unitarily invariant, that is, $d(U\rho U^\dagger, U\sigma U^\dagger) = d(\rho, \sigma)$.

We can use norms to find proper distances. One norm of particular interest is the Schatten p norm [23] of an operator X , given by $\|X\|_p := [\text{Tr}(|X|^p)]^{\frac{1}{p}}$, where p is a real number such that $p \geq 1$ and $|X| := \sqrt{X^\dagger X}$. The Schatten p norms of density operators ρ (i.e., $\rho \geq 0$, $\text{Tr}\rho = 1$, and $\rho^\dagger = \rho$) satisfy [23] unitary invariance, i.e., $\|U\rho V^\dagger\|_p = \|\rho\|_p$ for isometries U and V , and multiplicativity under tensor products, i.e., $\|\rho \otimes \sigma\|_p = \|\rho\|_p \|\sigma\|_p$.

Without further ado, the L_p distances d_p (also called *Schatten p distances*) [23,24] can be written as

$$d_p(\rho, \sigma) := \|\sigma - \rho\|_p = (\text{Tr}|\sigma - \rho|^p)^{\frac{1}{p}} \quad (12)$$

for all $p \geq 1$. For $p = 1$ and $p = 2$, the L_p distances are called the *trace distance* d_{Tr} and *Hilbert-Schmidt distance* d_{HS} , respectively. Although all the L_p distances are unitarily invariant because they inherit this property from the Schatten p norms, the only L_p distance that is contractive under CPTP maps is the trace distance [34]. However, all L_p distances are jointly convex,

$$d_p\left(\sum_i p_i \rho_i, \sum_i p_i \sigma_i\right) \leq \sum_i p_i d_p(\rho_i, \sigma_i). \quad (13)$$

Note that any power $p > 1$ of d_p is also jointly convex.

In addition, we have the *Bures distance* d_{Bu} [24,25],

$$d_{\text{Bu}}(\rho, \sigma) := [2 - 2\sqrt{F(\rho, \sigma)}]^{\frac{1}{2}}, \quad (14)$$

where $F(\rho, \sigma)$ is the *Uhlmann fidelity* [35] $F(\rho, \sigma) := \|\sqrt{\sigma}\sqrt{\rho}\|_1^2 = [\text{Tr}(\sqrt{\rho\sigma}\sqrt{\rho})^{\frac{1}{2}}]^2$, which is symmetric. Moreover, if $\rho = |\psi\rangle\langle\psi|$ and $\sigma = |\phi\rangle\langle\phi|$, then $F(\rho, \sigma) = |\langle\psi|\phi\rangle|^2$. Last, we have the *quantum Hellinger distance* d_{He} [24,26],

$$d_{\text{He}}(\rho, \sigma) := \|\sqrt{\sigma} - \sqrt{\rho}\|_2 = (2 - 2\text{Tr}\sqrt{\sigma}\sqrt{\rho})^{\frac{1}{2}}. \quad (15)$$

If ρ and σ commute, then $d_{\text{Bu}}(\rho, \sigma) = d_{\text{He}}(\rho, \sigma)$. Additional connections between the Hellinger distance, Bures distance, affinity, and fidelity are presented in Ref. [36]. Both the Bures and the Hellinger distances are contractive under CPTP maps. The joint convexity, however, is only satisfied by squaring them, i.e.,

$$d_{\text{Bu}}^2\left(\sum_i p_i \rho_i, \sum_i p_i \sigma_i\right) \leq \sum_i p_i d_{\text{Bu}}^2(\rho_i, \sigma_i), \quad (16)$$

with an identical expression for d_{He} [24]. A summary of the properties of all these distances is given in Table I.

III. RESULTS

A. Geometric conditional information

Let us propose a geometric measure of conditional information as

$$I^\square(\mathcal{E}|\mathcal{S})_\Omega := d_\square^n\left(\Omega, \Omega_S \otimes \frac{\mathbb{1}_\mathcal{E}}{d_\mathcal{E}}\right), \quad (17)$$

where $d_\square$ is any of the distances presented in Sec. IID and n is some power conveniently chosen. This definition is clearly

TABLE I. Summary of the properties that are satisfied by the trace distance d_{Tr} , the Hilbert-Schmidt distance d_{HS} , the L_p distances d_p with finite $p > 1$ and $p \neq 2$, the Bures distance d_{Bu} , the square Bures distance d_{Bu}^2 , the Hellinger distance d_{He} , and the square Hellinger distance d_{He}^2 between any pair of quantum states.

	d_{Tr}	d_{HS}	d_{HS}^2	d_p	d_p^p	d_{Bu}	d_{Bu}^2	d_{He}	d_{He}^2
Positive definiteness	✓	✓	✓	✓	✓	✓	✓	✓	✓
Unitary invariance	✓	✓	✓	✓	✓	✓	✓	✓	✓
Joint convexity	✓	✓	✓	✓	✓	✗	✓	✗	✓
Contractivity	✓	✗	✗	✗	✗	✓	✓	✓	✓

borrowed from the notion of conditional information with divergences that we presented in Sec. IIC and was inspired by the work of Roga *et al.* [26]. Immediately from the positive-definiteness property of distances, we have that

$$I^\square(\mathcal{E}|\mathcal{S})_\Omega = 0 \quad \text{iff} \quad \Omega = \Omega_S \otimes \frac{\mathbb{1}_\mathcal{E}}{d_\mathcal{E}}. \quad (18)$$

In addition, we can expect meaningful measures of conditional information from definition (17) once we use distances $d_\square^n$ that respect the contractivity relation (11), which is nothing more than a data-processing inequality in the geometrical context. This is the case for the trace distance and the square of the Bures and the Hellinger distances. As we are going to see, Hilbert-Schmidt distances can be contractive for unital maps.

B. Summary of main results: Geometric monotones of violations of quantum realism

In order to follow the realism-information relation (5), we must obtain the conditional information of the global states $\Omega_0 = \rho_S \otimes |e_0\rangle\langle e_0|$ and $\Omega_t = U_t(\Omega_0)U_t^\dagger$. The first one can be readily calculated using

$$I^\square(\mathcal{E}|\mathcal{S})_{\Omega_0} = d_\square^n\left(\rho \otimes |e_0\rangle\langle e_0|, \rho \otimes \frac{\mathbb{1}_\mathcal{E}}{d_\mathcal{E}}\right). \quad (19)$$

The conditional information of the global state after unitary interaction $U_{\mathcal{SE}}$ will be

$$I^\square(\mathcal{E}|\mathcal{S})_{\Omega_t} = d_\square^n\left(\Omega_t, \text{Tr}_\mathcal{E}(\Omega_t) \otimes \frac{\mathbb{1}_\mathcal{E}}{d_\mathcal{E}}\right). \quad (20)$$

Remember that the realism-information relation (5) comes from the fact that measurements establish realism and, therefore, $U_{\mathcal{SE}}$ must be such that

$$\text{Tr}_\mathcal{E}(\Omega_t) = \Phi_A(\rho_S). \quad (21)$$

As a result, the unitary $U_{\mathcal{SE}}$ satisfies (see Theorem 1 of Ref. [21])

$$U_{\mathcal{SE}}\left(\Phi_A(\rho_S) \otimes \frac{\mathbb{1}_\mathcal{E}}{d_\mathcal{E}}\right)U_{\mathcal{SE}}^\dagger = \Phi_A(\rho_S) \otimes \frac{\mathbb{1}_\mathcal{E}}{d_\mathcal{E}}, \quad (22)$$

with the condition that $d_{\mathcal{A}} = d_\mathcal{E}$. From the unitary invariance of distances and the above relation, we have

$$I^\square(\mathcal{E}|\mathcal{S})_{\Omega_t} = d_\square^n\left(\rho_S \otimes |e_0\rangle\langle e_0|, \Phi_A(\rho_S) \otimes \frac{\mathbb{1}_\mathcal{E}}{d_\mathcal{E}}\right). \quad (23)$$

Following (5), geometric monotones of VQRs (or simply monotones of realism) can be obtained using the following recipe:

$$\mathfrak{R}_A^\square(\rho_S) = \mathfrak{R}_A^{\max}(\square) - \Delta I^\square(\mathcal{E}|S)_{\Omega_0 \rightarrow \Omega_i}, \quad (24)$$

where $\Delta I^\square(\mathcal{E}|S)_{\Omega_0 \rightarrow \Omega_i} = I^\square(\mathcal{E}|S)_{\Omega_i} - I^\square(\mathcal{E}|S)_{\Omega_0}$ is the change in the geometric conditional information. Importantly, a *violation of quantum realism* is detected when $\mathfrak{R}_A^\square(\rho_S) < \mathfrak{R}_A^{\max}(\square)$. After some algebra (see Appendix B), we reach the following four geometric quantifiers of realism:

$$\mathfrak{R}_A^{\text{Tr}}(\rho_S) = \mathfrak{R}_A^{\max}(\text{Tr}) - \left[d_{\text{Tr}}\left(\rho_S, \frac{\Phi_A(\rho_S)}{d_{\mathcal{E}}}\right) - \frac{d_{\mathcal{E}} - 1}{d_{\mathcal{E}}} \right], \quad (25)$$

$$\mathfrak{R}_A^{\text{HS}}(\rho_S) = \mathfrak{R}_A^{\max}(\text{HS}) - \frac{1}{d_{\mathcal{E}}} d_{\text{HS}}^2(\rho_S, \Phi_A(\rho_S)), \quad (26)$$

$$\mathfrak{R}_A^{\text{Bu}}(\rho_S) = \mathfrak{R}_A^{\max}(\text{Bu}) - \frac{1}{\sqrt{d_{\mathcal{E}}}} d_{\text{Bu}}^2(\rho_S, \Phi_A(\rho_S)), \quad (27)$$

$$\mathfrak{R}_A^{\text{He}}(\rho_S) = \mathfrak{R}_A^{\max}(\text{He}) - \frac{1}{\sqrt{d_{\mathcal{E}}}} d_{\text{He}}^2(\rho_S, \Phi_A(\rho_S)), \quad (28)$$

where Tr, HS, Bu, and He indicate trace, Hilbert-Schmidt, Bures, and Hellinger. Remember that $d_{\mathcal{E}} = d_{\mathcal{A}}$. We have chosen $n = 1$ for the realism quantifier based on the trace distance and $n = 2$ for the other cases. The max value $\mathfrak{R}_A^{\max}(\square)$ can be obtained with optimization algorithms, but the analytical solution can be found by making ρ_S a maximally entangled state of two qudits in each case. In Fig. 1(a) we plot $\mathfrak{R}_A^{\max}$ for trace, Hilbert-Schmidt, Bures, and Hellinger distances as a function of the dimension $d_{\mathcal{E}}$ for a $d_{\mathcal{E}}$ -outcome observable $A = \{A_a\}_{a=0}^{d_{\mathcal{E}}-1}$ in the computational basis.

See the following example.

Example 1. Let us see how the realism of a spin-1/2 observable $A = \hat{v} \cdot \vec{\sigma}$ for the first particle of the Werner state

$$\rho_{\epsilon} = (1 - \epsilon) \frac{\mathbb{I}_4}{4} + \epsilon |\phi^+\rangle \langle \phi^+|, \quad (29)$$

where $|\phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, increases with the loss of coherence within the quantifiers mentioned above. The calculations were made with the help of symbolic computation software, and the lengthy expressions will be omitted, but the results are visualized in Fig. 1(b). As expected for the Werner state, the realism degree does not depend on the direction of measurement $\hat{v}$ because the maximally entangled state of two qubits is rotationally invariant. Notably, $\mathfrak{R}_A^{\text{Tr}}(\rho_{\epsilon})$ violates Axiom 1 for $\epsilon \in [0, 1/3]$, which makes the trace distance unsuitable for monotones of realism. It is also important to note that since ρ_{ϵ} commutes with $\Phi_A(\rho_{\epsilon})$, we have $\mathfrak{R}_A^{\text{Bu}}(\rho_{\epsilon}) = \mathfrak{R}_A^{\text{He}}(\rho_{\epsilon})$.

We can also propose to measure the realism degree of an observable by means of the L_p distances (12) for $p > 1$ and $p \neq 2$. Such a quantifier could generalize the trace and the Hilbert-Schmidt realities. However, the expression we have obtained for it does not provide any further insight [see Eq. (B26) in Appendix B]. Nonetheless, we present in Table II which axioms we know are fulfilled by $\mathfrak{R}_A^p$.

Another interesting aspect of quantifiers (25)–(28) induced by the geometric distances is that $\mathfrak{R}_A^{\max}(\square)$ falls asymptotically to zero with higher $d_{\mathcal{E}}$, eventually signaling full realism for any state and any observable. This behavior does not happen with entropic quantifiers such as (6), in which case

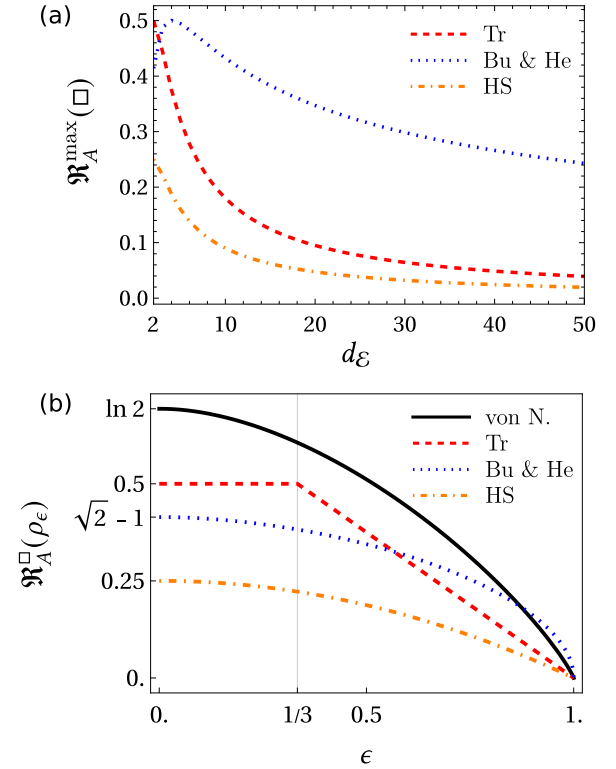


FIG. 1. (a) $\mathfrak{R}_A^{\max}$ for trace (red dashed line), Hilbert-Schmidt (orange dash-dotted line), and Bures and Hellinger (blue dotted line) distances as a function of the dimension $d_{\mathcal{E}}$. (b) Geometric quantifiers of VQRs $\mathfrak{R}_A^\square$ of any spin-1/2 observable regarding the first particle of a Werner state ρ_{ϵ} from Example 1 as a function of ϵ for von Neumann (6) (black line), trace (25) (red dashed line), Bures (27) and Hellinger (28) (blue dotted line), and Hilbert-Schmidt (26) (orange dash-dotted line) distances. $\mathfrak{R}_A^{\max}(\square)$ for $d_{\mathcal{E}} = 2$ is highlighted on the vertical axis.

we have $\mathfrak{R}_A^{\max}(\text{von N.}) = \ln d_{\mathcal{E}}$, where von. N indicates von Neumann.

Let us now investigate which of the four axioms mentioned in Sec. II B are satisfied by each one of the four geometric quantifiers of realism in Eqs. (25)–(28).

C. VQRs with trace distance

Example 1 alone is sufficient to disqualify $\mathfrak{R}_A^{\text{Tr}}$ as a *bona fide* monotone for VQRs, as it violates Axiom 1. Specif-

TABLE II. Summary of the axioms satisfied by the A -realism quantifiers $\mathfrak{R}_A^{\text{Tr}}$ [Eq. (25)], $\mathfrak{R}_A^{\text{HS}}$ [Eq. (26)], $\mathfrak{R}_A^p$, $\mathfrak{R}_A^{\text{Bu}}$ [Eq. (27)], and $\mathfrak{R}_A^{\text{He}}$ [Eq. (28)] built out of their corresponding distance measures d_{Tr} , d_{HS}^2 , d_p^p (for finite $p > 1$ and $p \neq 2$), d_{Bu}^2 , and d_{He}^2 , whose properties are listed in Table I. Our approach legitimizes just two geometric realism monotones, namely, $\mathfrak{R}_A^{\text{Bu}}$ and $\mathfrak{R}_A^{\text{He}}$.

	$\mathfrak{R}_A^{\text{Tr}}$	$\mathfrak{R}_A^{\text{HS}}$	$\mathfrak{R}_A^p$	$\mathfrak{R}_A^{\text{Bu}}$	$\mathfrak{R}_A^{\text{He}}$
Axiom 1 (measurements)	✗	✓	?	✓	✓
Axiom 2(a) (part discard)	✓	✓	?	✓	✓
Axiom 2(b) (uncorrelated part)	✓	✗	✗	✓	✓
Axiom 3 (uncertainty relation)	✗	✓	?	✓	✓
Axiom 4 (mixing)	✓	✓	✓	✓	✓

ically, maximum realism is achieved by states other than $\Phi_A(\rho)$. Indeed, the VQR quantifier based on the trace distance exhibits a sudden death of irreality for any spin-1/2 observable associated with either particle of the Werner state when $\epsilon \leq 1/3$ (see Fig. 1). Although the trace distance satisfies the property of positive definiteness, its application in Eq. (25) involves one entry that is non-normalized, namely, $\Phi_A(\rho_S)/d_E$. Consequently, deriving a VQR monotone via the realism-information relation (5) leads to a loss of Axioms 1 and 3. Nevertheless, it remains plausible to establish a quantum resource theory of irreality induced by the trace distance. In such a theory, even though states with $0 < \epsilon \leq 1/3$ violate $\Phi_A(\rho_\epsilon) = \rho_\epsilon$, they might lack sufficient resources to qualify as resourceful states. Nonetheless, the trace distance is invariant under the addition of an uncorrelated system, contractive under the discarding of parts of the system, and also jointly convex, therefore satisfying Axioms 2 and 4.

D. VQRs with Hilbert-Schmidt distance

Although the Hilbert-Schmidt distance is not contractive under CPTP maps in general, it becomes contractive under CPTP *unital* maps, that is, maps that keep the identity invariant. That is the case for projective measurements Φ_A , monitorings $\mathcal{M}_A^\epsilon$, and partial traces. If C is a CPTP unital map, then $\|C(\rho)\|_p \leq \|\rho\|_p$, where equality holds if ρ is a maximally mixed state. Also, equality holds if $p = 1$. In the particular case that C is a nonselective measurement (e.g., $C = \Phi_A$), then equality holds iff $\rho = C(\rho)$ (see Theorem 5.2 in Ref. [37]). In addition, the square of d_{HS} remains contractive because $x \mapsto x^2$ is an injective monotonically increasing function for $x \geq 0$. Therefore, the quantifier $\mathfrak{R}_A^{\text{HS}}$ in (26) satisfies Axiom 2(a). It is easy to see that Axioms 1, 3, and 4 are also valid given the positive definiteness and the joint convexity properties of the L_p distances. The irrelevance of the uncorrelated system [Axiom 2(b)], however, is not fulfilled because

$$\begin{aligned} d_{\text{HS}}^2(\rho \otimes \sigma, \Phi_A(\rho) \otimes \sigma) &= \|\rho \otimes \sigma - \Phi_A(\rho) \otimes \sigma\|_2^2 \\ &= \|\rho - \Phi_A(\rho)\|_2^2 \|\sigma\|_2^2 \\ &> d_{\text{HS}}^2(\rho, \Phi_A(\rho)) \end{aligned} \quad (30)$$

if σ is not pure. That means that a quantifier of VQRs induced by the Hilbert-Schmidt distance fails to ignore completely uncorrelated systems when assessing the degree of realism in a given context. In fact, this is a common issue regarding quantum correlations derived from the Hilbert-Schmidt distance, as argued by Piani [38] (see also [39]). Having said that, the quantifier $\mathfrak{R}_A^{\text{HS}}$ does not meet the minimum requirements to be considered a monotone of VQR.

E. VQR with Bures and Hellinger distances

When we try to derive quantifiers for VQR based on the realism-information relation (5), the Bures and Hellinger distances are the only ones—of those we have seen so far—that provide us with *bona fide* monotones, namely, $\mathfrak{R}_A^{\text{Bu}}$ and $\mathfrak{R}_A^{\text{He}}$ from Eqs. (27) and (28). This is due to the positive definiteness, contractivity, and joint convexity properties fulfilled by these distances. In addition, these distances have a close

connection to entropic quantities known as Rényi divergence [40],

$$D_\alpha(\rho||\sigma) := \frac{1}{\alpha - 1} \ln \frac{\text{Tr}(\rho^\alpha \sigma^{1-\alpha})}{\text{Tr} \rho} \quad (31)$$

for $\alpha \in (0, 1) \cup (1, +\infty)$, and sandwiched Rényi divergence [41,42],

$$\tilde{D}_\alpha(\rho||\sigma) := \frac{1}{\alpha - 1} \ln \left\{ \frac{1}{\text{Tr} \rho} \text{Tr} \left[\left(\sigma^{-\frac{1-\alpha}{2\alpha}} \rho \sigma^{-\frac{1-\alpha}{2\alpha}} \right)^\alpha \right] \right\}, \quad (32)$$

with the same conditions as quantity (31). Equations (31) and (32) are said to be generalizations of Eq. (7) because $D_{\alpha \rightarrow 1}(\rho||\sigma) = \tilde{D}_{\alpha \rightarrow 1}(\rho||\sigma) = S(\rho||\sigma)$. One can see that

$$d_{\text{Bu}}^2(\rho, \sigma) = 2 - 2 \exp \left(-\frac{1}{2} \tilde{D}_{\alpha=\frac{1}{2}}(\rho||\sigma) \right), \quad (33)$$

$$d_{\text{He}}^2(\rho, \sigma) = 2 - 2 \exp \left(-\frac{1}{2} D_{\alpha=\frac{1}{2}}(\rho||\sigma) \right), \quad (34)$$

which means that the Bures and Hellinger realism monotones can be written as injective increasing functions of special cases of the Rényi divergences (31) and (32), precisely when $\alpha = 1/2$. Since monotones of VQRs induced by Rényi and sandwiched Rényi divergences were successfully obtained in [21], we see that (33) and (34) also indicate that $\mathfrak{R}_A^{\text{Bu}}$ and $\mathfrak{R}_A^{\text{He}}$ satisfy the requirements to be framed as *bona fide* monotones of VQRs.

The main difference between the Bures and Hellinger distances is that the latter is sensitive to the noncommutativity of states. In fact, the role of this feature in the assessment of the VQR of observables is not clear to us—which suggests a path of research in line with the work of Martins *et al.* [43]. Nevertheless, the reader can get an idea of how close these quantifiers are to each other from the following example.

Example 2. Consider the two-qubit state

$$\rho_\mu = \frac{\mathbb{1}}{4} + \frac{\mu}{4}(\sigma_x \otimes \sigma_x - \sigma_y \otimes \sigma_y) + \frac{2\mu - 1}{4}\sigma_z \otimes \sigma_z, \quad (35)$$

where $\mu \in [0, 1]$. Particularly, $\rho_{\mu=1} = |\phi^+\rangle\langle\phi^+|$, where $|\phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, and $\rho_{\mu=0} = (|01\rangle\langle 01| + |10\rangle\langle 10|)/2$. Now, consider the observable

$$A = \hat{u} \cdot \vec{\sigma} = \cos \theta \sin \phi \sigma_x + \sin \theta \sin \phi \sigma_y + \cos \phi \sigma_z, \quad (36)$$

where θ and ϕ are the polar and azimuthal angles and $\vec{\sigma}$ is the Pauli vector. When applied to either qubit, map Φ_A results in a state $\Phi_A(\rho_\mu)$ that does not always commute with ρ_μ . After some algebra, one can see that $\mathfrak{R}_A^{\text{Bu}}(\rho_\mu)$ and $\mathfrak{R}_A^{\text{He}}(\rho_\mu)$ are dependent only on μ and ϕ . The slight differences between these two monotones of VQRs due to the noncommutativity between ρ_μ and $\Phi_A(\rho_\mu)$ can be seen in Fig. 2.

In Table II, we summarize our results with a list of axioms that a quantifier needs to satisfy to be framed as a *bona fide* monotone of VQRs.

IV. DISCUSSION

In this work, we have extended the framework for quantifying violations of quantum realism (also known as quantum irreality [1]) by exploring geometric distances as potential tools for constructing monotones of VQR. Geometric distances, due to their symmetry and computational simplicity, provide an intriguing alternative to relative entropies. Among

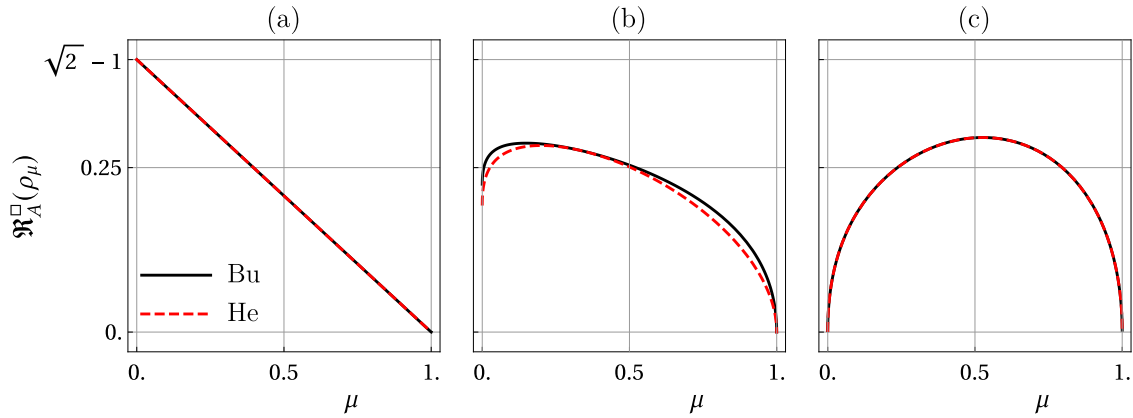


FIG. 2. Monotones of VQR $\mathfrak{R}_A^{\text{Bu}}$ (black solid line) and $\mathfrak{R}_A^{\text{He}}$ (red dashed line) of a spin-1/2 observable $A = \hat{v} \cdot \vec{\sigma}$, where $\hat{v} = (\cos \theta \sin \phi, \sin \theta \sin \phi, \cos \phi)$, for either qubit of the state ρ_μ (35) for (a) $\phi = 0$, (b) $\phi = \pi/4$, and (c) $\phi = \pi/2$.

the distances analyzed, we identified Bures and Hellinger distances as satisfying all minimal criteria for *bona fide* monotones of VQRs, while trace distance, Hilbert-Schmidt distance, and L_p distances in general fail to meet those criteria.

The realism-information relation (5) proposed in [21] imposes that quantifiers satisfying Axioms 1–4 for VQRs should inherently align with informational structures. Specifically, this relation sets a connection between conditional information (9) and the emergence of realism in quantum systems (20), therefore guiding the construction of monotones. The failure of certain distances, such as the trace distance and Hilbert-Schmidt distance, to meet the criteria for monotones stresses the requirements for valid quantifiers of VQRs. Specifically, the inability of these distances to consistently capture VQRs when $\Phi_A(\rho) \neq \rho$ [see the plateau in Fig. 1(b)] or their counterintuitive behavior under the addition of completely uncorrelated states [see (30)] highlights the importance of aligning distance-based monotones with physical principles of quantum realism.

Our findings suggest that quantifiers of VQRs that are not connected to entropic measures are not suitable for measuring violations of quantum realism since only the Bures and Hellinger distances could provide *bona fide* monotones of VQRs. Indeed, these distances can be expressed as functions of Rényi (34) and sandwiched Rényi divergences (33) in the specific cases when they are symmetric, which reinforces the connection of the Bures and Hellinger distances to informational measures.

Clearly, the set of distances we have explored in this work is not exhaustive. An intriguing open question is whether other geometric distances exist that satisfy all the axioms for valid VQR monotones. Identifying such distances could broaden the toolbox for quantifying violations of quantum realism and potentially reveal deeper connections between geometry and quantum information theory. Additionally, it remains to be seen whether these potential candidates share a fundamental link with entropic measures, similar to the Bures and Hellinger distances. Investigating this relationship could further elucidate the interplay between geometric and informational perspectives on quantum realism, offering new insights into the structure and interpretation of quantum measurements.

An important point to highlight is that $\mathfrak{R}_A^{\text{max}}(\square)$ for the trace, Hilbert-Schmidt, Hellinger, and Hellinger cases approaches zero as d_E increases, as shown in Fig. 1(a). This behavior contrasts with the original entropic measure of VQR given by (6), whose maximum value grows logarithmically with d_E . This observation raises several open questions: Is it essential for a measure of realism to assign higher values to observables in higher-dimensional systems? Should the realism-information relation (5) be rescaled to ensure that the geometric monotones exhibit increasing maximum values of realism as the dimension grows? These questions highlight potential directions for refining the theoretical framework of quantum realism.

In conclusion, this study advances the understanding of VQRs by identifying geometric distances that align with the informational principles underlying quantum realism. Our results provide both theoretical insights and practical tools for analyzing the interplay between measurement, information, and notions of realism in quantum mechanics, offering a promising avenue for future research in quantum foundations and information science.

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DATA AVAILABILITY

No data were created or analyzed in this study.

APPENDIX A: AUXILIARY LEMMAS

The proof of the following lemma can be found in Appendix A of [44].

Lemma A1. For any function f , any quantum states ρ and σ , and any observable A , we have

$$\text{Tr}[\rho f(\Phi_A(\sigma))] = \text{Tr}[\Phi_A(\rho) f(\Phi_A(\sigma))], \quad (\text{A1})$$

where Φ_A is the map of nonselective projective measurements given by $\Phi_A(\rho) = \sum_a (A_a \otimes \mathbb{1}_{\mathcal{B}}) \rho (A_a \otimes \mathbb{1}_{\mathcal{B}})$, where $A_a = |a\rangle\langle a|$ are projectors that sum up to identity in $\mathcal{H}_{\mathcal{A}}$.

Lemma A2. The Hilbert-Schmidt norm $\|\cdot\|_2 = \sqrt{\text{Tr}|\cdot|^2}$ satisfies

$$\|\rho\|_2^2 - \|\Phi_A(\rho)\|_2^2 = \|\rho - \Phi_A(\rho)\|_2^2 \quad (\text{A2})$$

for any state ρ and any nonselective projective measurement map Φ_A .

Proof. The proof relies in the direct application of Lemma A1:

$$\|\rho - \Phi_A(\rho)\|_2^2 = \text{Tr}[\rho - \Phi_A(\rho)]^2 \quad (\text{A3})$$

$$= \text{Tr}\rho^2 - 2\text{Tr}\rho\Phi_A(\rho) + \text{Tr}\Phi_A(\rho)^2 \quad (\text{A4})$$

$$= \text{Tr}\rho^2 - 2\text{Tr}\rho\Phi_A(\rho) + \text{Tr}\rho\Phi_A(\rho) \quad (\text{Lemma A1}) \quad (\text{A5})$$

$$= \text{Tr}\rho^2 - \text{Tr}\rho\Phi_A(\rho) \quad (\text{A6})$$

$$= \text{Tr}\rho^2 - \text{Tr}\Phi_A(\rho)^2 \quad (\text{Lemma A1}) \quad (\text{A7})$$

$$= \|\rho\|_2^2 - \|\Phi_A(\rho)\|_2^2. \quad (\text{A8})$$

APPENDIX B: CALCULATIONS

For $\Omega_0 = \rho \otimes \mathbb{1}_{\mathcal{E}}/d_{\mathcal{E}}$, Eq. (17) reads

$$I_{\mathcal{E}|S}^{\square}(\Omega_0) = d_{\square}^n \left(\Omega_0, \text{Tr}_{\mathcal{E}}\Omega_0 \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}} \right) \quad (\text{B1})$$

$$= d_{\square}^n \left(\rho \otimes |e_0\rangle\langle e_0|, \rho \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}} \right). \quad (\text{B2})$$

After the unitary evolution, we have

$$I_{\mathcal{E}|S}^{\square}(\Omega_t) = d_{\square}^n \left(\Omega_t, \text{Tr}_{\mathcal{E}}\Omega_t \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}} \right) \quad (\text{B3})$$

$$= d_{\square}^n \left(U_t(\rho \otimes |e_0\rangle\langle e_0|)U_t^{\dagger}, \Phi_A(\rho) \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}} \right) \quad (\text{B4})$$

$$= d_{\square}^n (U_t(\rho \otimes |e_0\rangle\langle e_0|)U_t^{\dagger}, U_t \left(\Phi_A(\rho) \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}} \right) U_t^{\dagger}) \quad (\text{B5})$$

$$= d_{\square}^n \left(\rho \otimes |e_0\rangle\langle e_0|, \Phi_A(\rho) \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}} \right). \quad (\text{B6})$$

Note that we have used (22) in (B5). Now, let us evaluate the above expressions for the following specific cases.

1. L_p distances

The L_p distance is defined as [23,24]

$$d_p(\rho, \sigma) := \|\sigma - \rho\|_p = (\text{Tr}|\sigma - \rho|^p)^{\frac{1}{p}} \quad (\text{B7})$$

for all $p \geq 1$. Therefore, (B2) becomes

$$I_{\mathcal{E}|S}^p(\Omega_0) = d_p^p \left(\rho \otimes |e_0\rangle\langle e_0|, \rho \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}} \right) \quad (\text{B8})$$

$$= \left\| \rho \otimes |e_0\rangle\langle e_0| - \rho \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}} \right\|_p^p \quad (\text{B9})$$

$$= \|\rho\|_p^p \left\| |e_0\rangle\langle e_0| - \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}} \right\|_p^p \quad (\text{B10})$$

$$= \|\rho\|_p^p \left\| \sum_{k=0}^{d_{\mathcal{E}}-1} \left(\delta_{k,0} - \frac{1}{d_{\mathcal{E}}} \right) |e_k\rangle\langle e_k| \right\|_p^p \quad (\text{B11})$$

$$= \|\rho\|_p^p \left\| \text{Tr} \left[\sum_{k=0}^{d_{\mathcal{E}}-1} \left(\delta_{k,0} - \frac{1}{d_{\mathcal{E}}} \right) |e_k\rangle\langle e_k| \right] \right\|_p^p \quad (\text{B12})$$

$$= \|\rho\|_p^p \left\| \sum_{k=0}^{d_{\mathcal{E}}-1} \left| \delta_{k,0} - \frac{1}{d_{\mathcal{E}}} \right|^p \text{Tr}|e_k\rangle\langle e_k| \right\|_p^p \quad (\text{B13})$$

$$= \|\rho\|_p^p \left\| \sum_{k=0}^{d_{\mathcal{E}}-1} \left| \delta_{k,0} - \frac{1}{d_{\mathcal{E}}} \right|^p \right\|_p^p \quad (\text{B14})$$

$$= \|\rho\|_p^p \left\| \left(1 - \frac{1}{d_{\mathcal{E}}} \right)^p + \frac{d_{\mathcal{E}} - 1}{d_{\mathcal{E}}^p} \right\|_p^p \quad (\text{B15})$$

$$= \|\rho\|_p^p \frac{1}{d_{\mathcal{E}}^p} [(d_{\mathcal{E}} - 1)^p + d_{\mathcal{E}} - 1]. \quad (\text{B16})$$

Similarly, for (B6) we have

$$I_{\mathcal{E}|S}^p(\Omega_t) = d_p^p \left(\rho \otimes |e_0\rangle\langle e_0|, \Phi_A(\rho) \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}} \right) \quad (\text{B17})$$

$$= \left\| \rho \otimes |e_0\rangle\langle e_0| - \Phi_A(\rho) \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}} \right\|_p^p \quad (\text{B18})$$

$$= \left\| \sum_{k=0}^{d_{\mathcal{E}}-1} \left(\delta_{k,0} \rho - \frac{\Phi_A(\rho)}{d_{\mathcal{E}}} \right) \otimes |e_k\rangle\langle e_k| \right\|_p^p \quad (\text{B19})$$

$$= \text{Tr} \left| \sum_{k=0}^{d_{\mathcal{E}}-1} \left(\delta_{k,0} \rho - \frac{\Phi_A(\rho)}{d_{\mathcal{E}}} \right) \otimes |e_k\rangle\langle e_k| \right|^p \quad (\text{B20})$$

$$= \sum_{k=0}^{d_{\mathcal{E}}-1} \text{Tr} \left| \delta_{k,0} \rho - \frac{\Phi_A(\rho)}{d_{\mathcal{E}}} \right|^p \text{Tr}|e_k\rangle\langle e_k| \quad (\text{B21})$$

$$= \sum_{k=0}^{d_{\mathcal{E}}-1} \text{Tr} \left| \delta_{k,0} \rho - \frac{\Phi_A(\rho)}{d_{\mathcal{E}}} \right|^p \quad (\text{B22})$$

$$= \text{Tr} \left| \rho - \frac{\Phi_A(\rho)}{d_{\mathcal{E}}} \right|^p + (d_{\mathcal{E}} - 1) \text{Tr} \left| \frac{\Phi_A(\rho)}{d_{\mathcal{E}}} \right|^p \quad (\text{B23})$$

$$= d_p^p \left(\rho, \frac{\Phi_A(\rho)}{d_{\mathcal{E}}} \right) + \frac{d_{\mathcal{E}} - 1}{d_{\mathcal{E}}^p} \|\Phi_A(\rho)\|_p^p. \quad (\text{B24})$$

Finally, the change in conditional information induced by the L_p distances is given by

$$\Delta I_{\mathcal{E}|S}^p(\Omega_t, \Omega_0) = d_p^p \left(\rho, \frac{\Phi_A(\rho)}{d_{\mathcal{E}}} \right) - \frac{d_{\mathcal{E}} - 1}{d_{\mathcal{E}}^p} (\|\Phi_A(\rho)\|_p^p - \|\rho\|_p^p) \quad (\text{B25})$$

$$- \|\rho\|_p^p \left(\frac{d_{\mathcal{E}} - 1}{d_{\mathcal{E}}} \right)^p. \quad (\text{B26})$$

In what follows, we derive expressions for $\Delta I_{\mathcal{E}|S}^{\text{Tr}}(\Omega_t, \Omega_0)$ for the trace distance ($p = 1$),

$$I_{\mathcal{E}|S}^{\text{Tr}}(\Omega_0) = 2 \left(\frac{d_{\mathcal{E}} - 1}{d_{\mathcal{E}}} \right), \quad (\text{B27})$$

$$I_{\mathcal{E}|S}^{\text{Tr}}(\Omega_t) = d_{\text{Tr}} \left(\rho, \frac{\Phi_A(\rho)}{d_{\mathcal{E}}} \right) + \frac{d_{\mathcal{E}} - 1}{d_{\mathcal{E}}}, \quad (\text{B28})$$

$$\Delta I_{\mathcal{E}|S}^{\text{Tr}}(\Omega_t, \Omega_0) = d_{\text{Tr}} \left(\rho, \frac{\Phi_A(\rho)}{d_{\mathcal{E}}} \right) - \frac{d_{\mathcal{E}} - 1}{d_{\mathcal{E}}}, \quad (\text{B29})$$

and the Hilbert-Schmidt distance ($p = 2$),

$$I_{\mathcal{E}|S}^{\text{HS}}(\Omega_0) = \frac{d_{\mathcal{E}} - 1}{d_{\mathcal{E}}} \|\rho\|_2^2, \quad (\text{B30})$$

$$I_{\mathcal{E}|S}^{\text{HS}}(\Omega_t) = \text{Tr} \left(\rho - \frac{\Phi_A(\rho)}{d_{\mathcal{E}}} \right)^2 + \frac{d_{\mathcal{E}} - 1}{d_{\mathcal{E}}^2} \text{Tr} \Phi_A(\rho)^2 \quad (\text{B31})$$

$$\begin{aligned} &= \text{Tr} \rho^2 - \frac{2}{d_{\mathcal{E}}} \text{Tr} \rho \Phi_A(\rho) \\ &\quad + \frac{1}{d_{\mathcal{E}}^2} \text{Tr} \Phi_A(\rho)^2 + \frac{d_{\mathcal{E}} - 1}{d_{\mathcal{E}}^2} \text{Tr} \Phi_A(\rho)^2 \end{aligned} \quad (\text{B32})$$

$$= \text{Tr} \rho^2 - \frac{2}{d_{\mathcal{E}}} \text{Tr} \Phi_A(\rho)^2 + \frac{1}{d_{\mathcal{E}}} \text{Tr} \Phi_A(\rho)^2 \quad (\text{B33})$$

$$= \|\rho\|_2^2 - \frac{1}{d_{\mathcal{E}}} \|\Phi_A(\rho)\|_2^2, \quad (\text{B34})$$

$$\Delta I_{\mathcal{E}|S}^{\text{HS}}(\Omega_t, \Omega_0) = \frac{1}{d_{\mathcal{E}}} [\|\rho\|_2^2 - \|\Phi_A(\rho)\|_2^2] \quad (\text{B35})$$

$$= \frac{1}{d_{\mathcal{E}}} \|\rho - \Phi_A(\rho)\|_2^2 \quad (\text{B36})$$

$$= \frac{1}{d_{\mathcal{E}}} d_{\text{HS}}^2(\rho, \Phi_A(\rho)). \quad (\text{B37})$$

Note that we used Lemma A 2 in Eqs. (B33) and (B36).

2. Bures distance

For the Bures distance, first, we need the Uhlmann fidelity [35] given by $F(\rho, \sigma) := \|\sqrt{\sigma} \sqrt{\rho}\|_1^2 = [\text{Tr}(\sqrt{\rho} \sigma \sqrt{\rho})^{\frac{1}{2}}]^2$. The fidelity between $\rho \otimes |e_0\rangle\langle e_0|$ and $\rho \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}}$ is

$$\begin{aligned} &F \left(\rho \otimes |e_0\rangle\langle e_0|, \rho \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}} \right) \\ &= \left\| \sqrt{\rho \otimes |e_0\rangle\langle e_0|} \sqrt{\rho \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}}} \right\|_1^2 \end{aligned} \quad (\text{B38})$$

$$= \left\| (\sqrt{\rho} \otimes |e_0\rangle\langle e_0|) \left(\sqrt{\rho} \otimes \frac{\mathbb{1}_{\mathcal{E}}}{\sqrt{d_{\mathcal{E}}}} \right) \right\|_1^2 \quad (\text{B39})$$

$$= \left\| \rho \otimes \frac{|e_0\rangle\langle e_0|}{\sqrt{d_{\mathcal{E}}}} \right\|_1^2 \quad (\text{B40})$$

$$= \frac{1}{d_{\mathcal{E}}} \|\rho\|_1^2 \| |e_0\rangle\langle e_0| \|_1^2 \quad (\text{B41})$$

$$= \frac{1}{d_{\mathcal{E}}}. \quad (\text{B42})$$

Similarly, the fidelity between $\rho \otimes |e_0\rangle\langle e_0|$ and $\Phi_A(\rho) \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}}$ can be calculated as

$$\begin{aligned} &F \left(\rho \otimes |e_0\rangle\langle e_0|, \Phi_A(\rho) \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}} \right) \\ &= \left\| \sqrt{\rho \otimes |e_0\rangle\langle e_0|} \sqrt{\Phi_A(\rho) \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}}} \right\|_1^2 \end{aligned} \quad (\text{B43})$$

$$= \left\| (\sqrt{\rho} \otimes |e_0\rangle\langle e_0|) \left[\sqrt{\Phi_A(\rho)} \otimes \frac{\mathbb{1}_{\mathcal{E}}}{\sqrt{d_{\mathcal{E}}}} \right] \right\|_1^2 \quad (\text{B44})$$

$$= \left\| \sqrt{\rho} \sqrt{\Phi_A(\rho)} \otimes \frac{|e_0\rangle\langle e_0|}{\sqrt{d_{\mathcal{E}}}} \right\|_1^2 \quad (\text{B45})$$

$$= \frac{1}{d_{\mathcal{E}}} \left\| \sqrt{\rho} \sqrt{\Phi_A(\rho)} \right\|_1^2 \| |e_0\rangle\langle e_0| \|_1^2 \quad (\text{B46})$$

$$= \frac{1}{d_{\mathcal{E}}} F(\rho, \Phi_A(\rho)). \quad (\text{B47})$$

Now, we can calculate

$$I_{\mathcal{E}|S}^{\text{Bu}}(\Omega_0) = d_{\text{Bu}}^2 \left(\rho \otimes |e_0\rangle\langle e_0|, \rho \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}} \right) \quad (\text{B48})$$

$$= 2 - 2 \sqrt{F \left(\rho \otimes |e_0\rangle\langle e_0|, \rho \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}} \right)} \quad (\text{B49})$$

$$= 2 - \frac{2}{\sqrt{d_{\mathcal{E}}}}, \quad (\text{B50})$$

$$I_{\mathcal{E}|S}^{\text{Bu}}(\Omega_t) = d_{\text{Bu}}^2 \left(\rho \otimes |e_0\rangle\langle e_0|, \Phi_A(\rho) \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}} \right) \quad (\text{B51})$$

$$= 2 - 2 \sqrt{F \left(\rho \otimes |e_0\rangle\langle e_0|, \Phi_A(\rho) \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}} \right)} \quad (\text{B52})$$

$$= 2 - 2 \sqrt{\frac{1}{d_{\mathcal{E}}} F(\rho, \Phi_A(\rho))}, \quad (\text{B53})$$

$$\Delta I_{\mathcal{E}|S}^{\text{Bu}}(\Omega_t, \Omega_0) = \frac{1}{\sqrt{d_{\mathcal{E}}}} [2 - 2 \sqrt{F(\rho, \Phi_A(\rho))}] \quad (\text{B54})$$

$$= \frac{1}{\sqrt{d_{\mathcal{E}}}} d_{\text{Bu}}^2(\rho, \Phi_A(\rho)). \quad (\text{B55})$$

3. Hellinger distance

The quantum Hellinger distance d_{He} [24,26] is given by $d_{\text{He}}(\rho, \sigma) := \|\sqrt{\sigma} - \sqrt{\rho}\|_2 = (2 - 2\text{Tr} \sqrt{\sigma} \sqrt{\rho})^{\frac{1}{2}}$. Therefore,

$$I_{\mathcal{E}|S}^{\text{He}}(\Omega_0) = d_{\text{He}}^2 \left(\rho \otimes |e_0\rangle\langle e_0|, \rho \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}} \right) \quad (\text{B56})$$

$$= 2 - 2\text{Tr} \left(\sqrt{\rho \otimes |e_0\rangle\langle e_0|} \sqrt{\rho \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}}} \right) \quad (\text{B57})$$

$$= 2 - 2\text{Tr} \left(\rho \otimes \frac{|e_0\rangle\langle e_0|}{\sqrt{d_{\mathcal{E}}}} \right) \quad (\text{B58})$$

$$= 2 - \frac{2}{\sqrt{d_{\mathcal{E}}}} \text{Tr}(\rho) \text{Tr}(|e_0\rangle\langle e_0|) \quad (\text{B59})$$

$$= 2 - \frac{2}{\sqrt{d_{\mathcal{E}}}}, \quad (\text{B60})$$

$$I_{\mathcal{E}|S}^{\text{He}}(\Omega_t) = d_{\text{He}}^2 \left(\rho \otimes |e_0\rangle\langle e_0|, \Phi_A(\rho) \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}} \right) \quad (\text{B61})$$

$$= 2 - 2\text{Tr} \left(\sqrt{\rho \otimes |e_0\rangle\langle e_0|} \sqrt{\Phi_A(\rho) \otimes \frac{\mathbb{1}_{\mathcal{E}}}{d_{\mathcal{E}}}} \right) \quad (\text{B62})$$

$$= 2 - 2\text{Tr} \left(\sqrt{\rho} \sqrt{\Phi_A(\rho)} \otimes \frac{|e_0\rangle\langle e_0|}{\sqrt{d_{\mathcal{E}}}} \right) \quad (\text{B63})$$

$$= 2 - \frac{2}{\sqrt{d_{\mathcal{E}}}} \text{Tr}[\sqrt{\rho} \sqrt{\Phi_A(\rho)}] \text{Tr}(|e_0\rangle\langle e_0|) \quad (\text{B64})$$

$$= 2 - \frac{2}{\sqrt{d_{\mathcal{E}}}} \text{Tr}[\sqrt{\rho} \sqrt{\Phi_A(\rho)}], \quad (\text{B65})$$

$$\Delta I_{\mathcal{E}|S}^{\text{He}}(\Omega_t, \Omega_0) = \frac{1}{\sqrt{d_{\mathcal{E}}}} \{2 - 2\text{Tr}[\sqrt{\rho} \sqrt{\Phi_A(\rho)}]\} \quad (\text{B66})$$

$$= \frac{1}{\sqrt{d_{\mathcal{E}}}} d_{\text{He}}^2(\rho, \Phi_A(\rho)). \quad (\text{B67})$$

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